HUDSON C. SMITH

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EXPERIENCE

GrayMatter Robotics — *Senior Robotics Engineer — Government Projects* Los Angeles, CA | Aug 2026 – Present

- Full-scale project lead and mentor for the government projects team: trained 2 interns through basic onboarding and serve as the primary technical screening interviewer for new team hires.
- Customer-facing technical lead for government NRE programs in robotic grinding, ultrasonic inspection, and cold spray repair
- Source opportunities, write proposals, scope requirements, execute R&D (hands-on developer leading a technical team of software and applications engineers), and own the customer relationship through program execution.
- Serve as a company delegate to the American Foundry Society (AFS); build relationships with defense and manufacturing decision-makers and advance adoption of robotic automation

GrayMatter Robotics — *Robotics Engineer II — Government Projects* Los Angeles, CA | Jul 2025 – Aug 2026

- Performed hands-on systems engineering, hardware/software debugging, rapid prototyping, and coordination to prove out customer needs internally before looping in software and mechatronics resources. Delivered final R&D results as fully developed demos, technical reports, pilot programs, and deployment-grade systems
- Supported proposal development and technical scoping for new government NRE opportunities, translating customer requirements into R&D execution plans and reporting deliverables.

GrayMatter Robotics — *Robotics Engineer — Systems & Applications* Los Angeles, CA | Apr 2024 – Jul 2025

- Took a commercial aerospace MRO robotic system from R&D into production as primary on-site engineer: developed scanning poses/parameters, tuned planning and software configs, wrote SOPs, trained customer personnel, and handed off a live production system as a retained account.
- Led scan-to-path abrasive cutting and belt grinding capabilities from scratch with tooling (PushCorp) and abrasive (3M) partners: devised process parameters, authored software requirements, and drove implementation through customer demonstrations.
- Led on-site commissioning and live troubleshooting of cement truck refurbishment robotic system through customer handoff
- Conducted technical site assessments at prospective customer facilities to evaluate automation feasibility, value proposition, surface constraints, and define solution architectures for new programs

ABB Corporate Research — *Robotics Research Engineer (Working Student)* Ladenburg, Germany | May 2023 – Nov 2023

- Applied ML to predict safe stopping distances for industrial robots under ISO/TS 15066 and ISO 10218 - designed and executed experiments on ABB IRB-1100 robot across speeds and payloads, collected over 30,000 unique data entries for robot stopping trajectories, built probabilistic safety models using PyTorch from empirical data.
- Published findings at VDI Mechatroniktagung; work constituted the basis of master's thesis: Safe Stopping Distances and Times for Industrial Robotics

SELECTED PROGRAMS

Robotic Plate Grinding & Ultrasonic Inspection — Defense Manufacturer | NRE

- Technical lead from Request for Proposal through R&D execution; integrated industrial robot, 3D vision, ultrasonic sensor and force-controlled tooling; managed government stakeholder relationship end-to-end

Aerospace MRO Robotic Automation — Commercial Airline | Deployed

- Lead on-site engineer who took the system from R&D into production: devised scanning and process parameters, tuned planning configs, wrote SOPs, trained all customer personnel; system remains live under active sustainment contract

Scan-to-Robot Abrasive Cutting Pipeline — Leading Global Producer of Industrial Machinery | R&D Demonstration

- On-site technical discovery with tooling partner → authored software requirements → co-developed with engineering team through working customer demonstration; capability served as proof point for subsequent contract award

Safe Stopping Distance Models — ABB Corporate Research | Published

- Probabilistic ML models for robot stopping behavior under ISO 10218/TS 15066 for dual-channel, safety-rated stopping distance and time modeling; experimentally validated on ABB systems; published at VDI Mechatroniktagung

TECHNICAL SKILLS

Programming: Python (PyTorch, NumPy, OpenCV, Matplotlib), ROS 2, MATLAB, C++, C, VBA

Robotics: Industrial robot programming (ABB, FANUC), robotic grinding & surface finishing, scan-to-path automation, force-controlled manipulation, closed-loop control logic, high-mix and MRO system design

Vision & Perception: 3D scanning, point cloud processing (segmentation), machine vision integration, depth sensor fusion

ML / AI: Supervised learning, regression & classification, experimental design, safety-critical ML system design

Engineering Tools: NVIDIA Isaac Sim, MeshLab, Autodesk Revit, AutoCAD, SolidWorks, ABB RobotStudio, FANUC RoboGuide

EDUCATION

Virginia Tech - *M.S. Mechanical Engineering* 2023

Technische Universität Darmstadt - *M.Sc. Mechanical & Process Engineering* 2023

Hamilton College - *B.A. Physics & German Studies* 2021

HONORS & PUBLICATIONS

- Deutschlandstipendium Scholarship - TU Darmstadt (merit-based, German Federal Government)
- Fulbright ETA Fellowship - Alternate, Germany
- Nottingham Leadership Scholarship - Delta Phi Chi Chapter, Hamilton College (merit-based)
- Graduate Teaching Assistant - Virginia Tech Physics Department
- Publication: Smith, “Prediction of Stopping Behavior for Robot Safety,” (First Author, VDI Mechatroniktagung, 2023)
- Publication: “Quantifying Squaraine Phase Separation, Aggregate Populations, and Solar Cell Performance with a Correlated Multimethod Approach, (Journal of Physical Chemistry C, 2022)